

Act III: Casting the Sellers

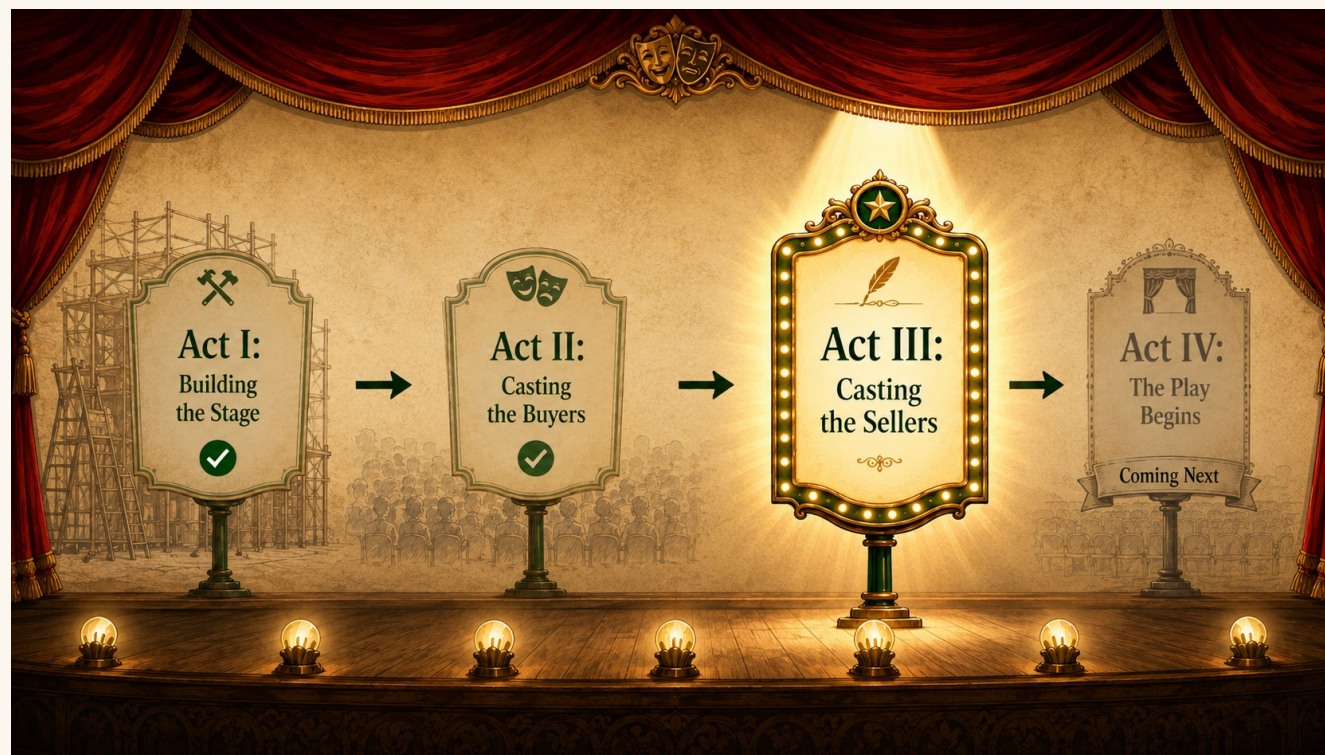
Production, Costs and Supply Decisions

Module VI: Backstage at the Bakery

Production, Technology, and Costs



Previously in Marginal Falls



Next: Sellers' side of the story

The Other Side of the Story

Buyers may want daily bread, festival cakes, ferry rides, or snacks.
But someone must produce them.



How do sellers decide what they can make?

The Festival Order Arrives

Marginal Falls is hosting the Founders' Night Block Party.

The Bakery is asked to prepare:

- Daily bread
- Pastries
- Festival cakes
- Snack boxes for the town square



Can the bakery handle the order?

Role of a Firm

A firm is an organization that uses inputs in a **technology** to produce goods and services.



Marginal Falls examples:

- Bakery
- Ferry company
- Coffee shop
- Repair shop
- Drone & Forensics Lab

Technology

Technology is the process a firm uses to turn inputs into outputs.

Technology is not only computers.

It can be a recipe, oven system, kitchen layout, schedule, or production routine.

The bakery's **technology** is how they use flour, labor, and ovens into baked goods.

Technological Change

Technological change is a change in the firm's ability to produce output from the same inputs.

- **Positive technological change:**
More output from the same inputs.
- **Negative technological change:**
Less output from the same inputs.



What Limits the Bakery Tonight?

The bakery can hire extra workers.

It can buy more flour.

It can pay for more packaging.

But it cannot instantly build a bigger kitchen.



Which inputs can change now, and which cannot?

Short Run Vs Long Run

The short run is the period during which at least one input is fixed.



Tonight, the bakery has only 2 ovens. It can hire more workers, but it cannot add a third oven immediately.

The long run is the period in which all inputs can be changed.



Next month, the bakery can rent a bigger kitchen, buy more ovens, change its layout, or open another branch.

Party Invitation

econgames.promitkchaudhuri.com/party/

Goal: Choose a party size that keeps the cost per guest reasonable.

Watch carefully:

- What happens when only a few guests attend?
- What happens when the party gets crowded?



Fixed Cost

Fixed cost is a cost that does not change as output changes.



Party example:

The park permit must be paid whether 20 or 100 people attend.

Bakery example:

The price for ovens must be paid whether the bakery makes 10 or 100 pastries.

Variable Cost

Variable cost is a cost that changes as output changes.



Party example:

More guests require more food, drinks, plates, and cleanup.

Bakery example:

More pastries require more flour, sugar, packaging, and labor hours.

Total Cost

Total cost is the cost of all inputs used in production.

Total Cost = Fixed Cost + Variable Cost

$$TC = FC + VC$$

Marginal Falls Party:

Total party cost = setup cost + guest-related cost.

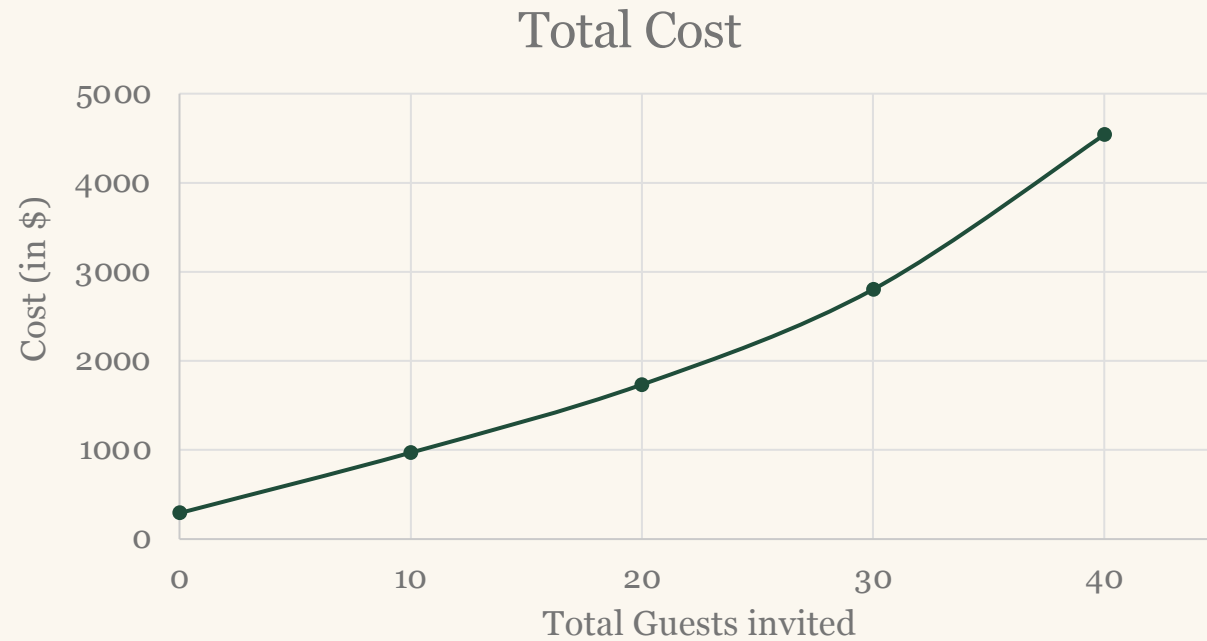
The Party Cost Ledger

For 4 people
organizing the
party and
inviting guests.

Guests	Fixed Cost	Variable Cost	Total Cost
0	\$292	\$0	\$292
10	\$292	\$678	\$970
20	\$292	\$1,442	\$1,734
30	\$292	\$2,510	\$2,802
40	\$292	\$4,247	\$4,539
50	\$292	\$7,164	\$7,456

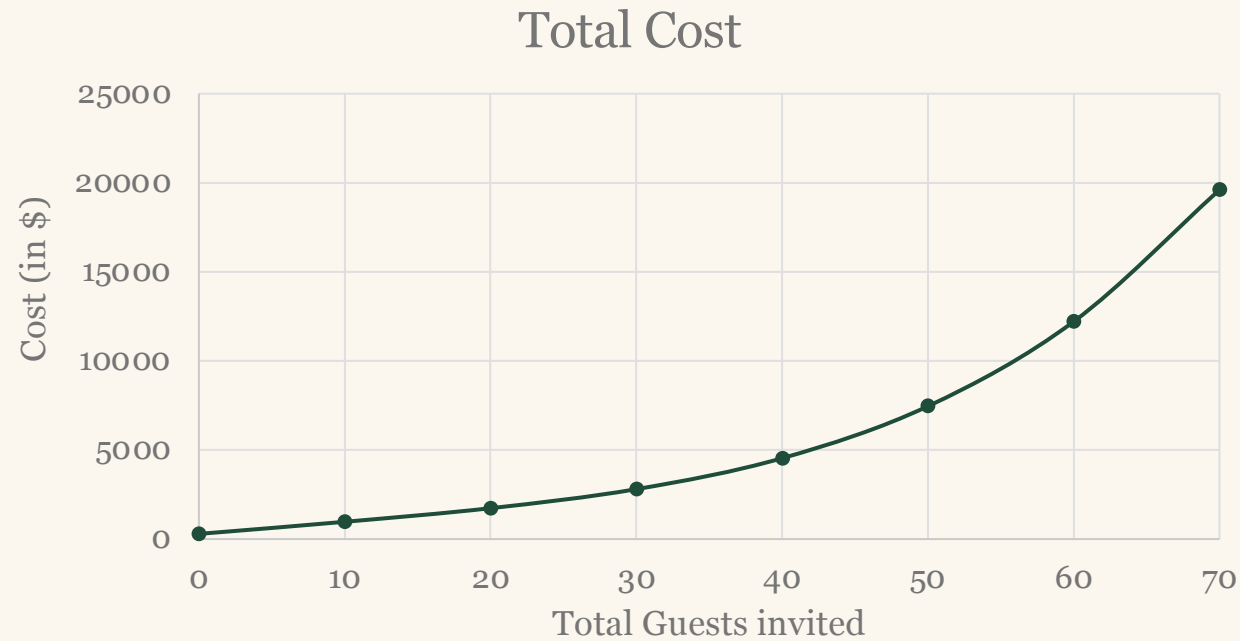
Fixed cost stays the same. Variable cost rises. Total cost rises.

Total Cost Curve



Total cost starts above zero because fixed cost exists.
Even if there's no guest, the rent for the local park must be paid in advance.

Total Cost Curve



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Quiz Time

The short run is the period during which at least one input is fixed.

The long run is the period in which all inputs can be changed.



What is the FC and VC in Long Run?

Average Total Cost

Average total cost is total cost divided by output.

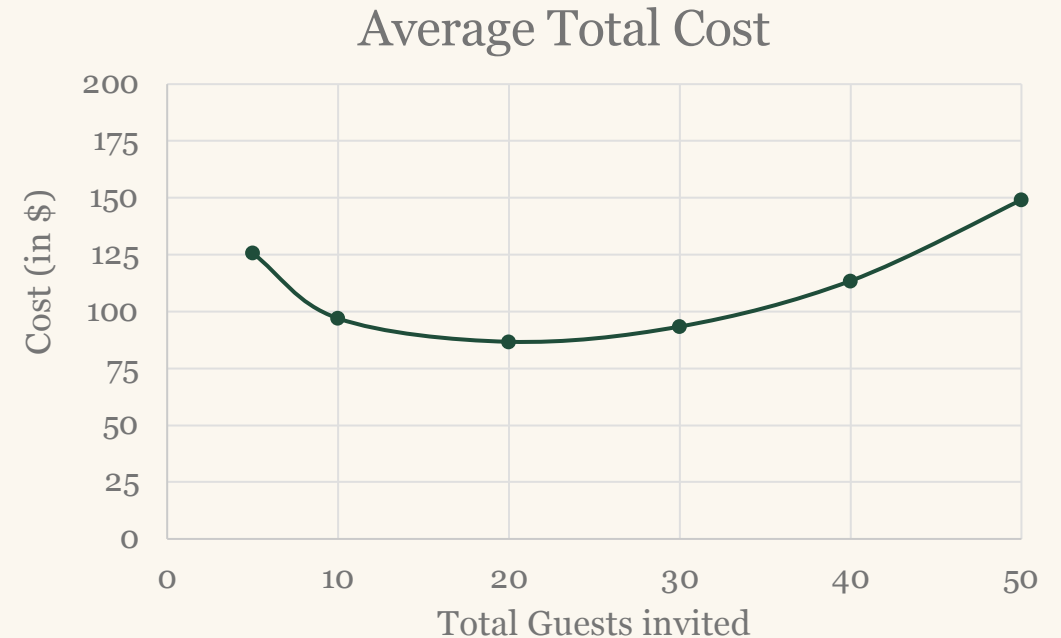
$$ATC = TC / Q$$

Marginal Falls Party:

Cost Split per Person

Cost Split per Person

Guests	Total Cost	Average Total Cost
10	\$970	\$97
20	\$1,734	\$86.70
30	\$2,802	\$93.40
40	\$4,539	\$113.48
50	\$7,456	\$149.12



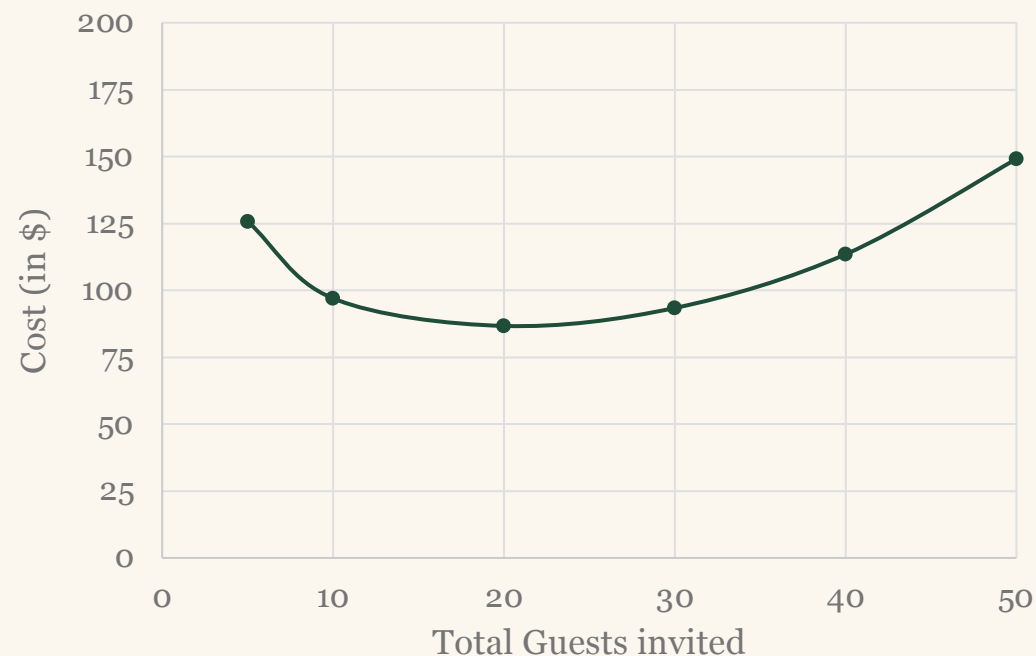
Average total cost falls first, then rises.

Why Did the Cost Split Fall First?

At 10 guests, the cost split is high.
At 20 guests, the cost split is lower.

But the party did not become cheaper
overall.

Total cost increased.



How can total cost rise while the cost split falls?

Spreading the Park Rent

The park rent is fixed.

It must be paid whether the party is small or large.
When more guests attend, the same **Fixed Cost** is spread across more people.

More people sharing the same park rent bill can make the split cheaper.

Average Fixed Cost

Remember:

$$TC = FC + VC$$

Average Fixed Cost is Fixed cost divided by output.

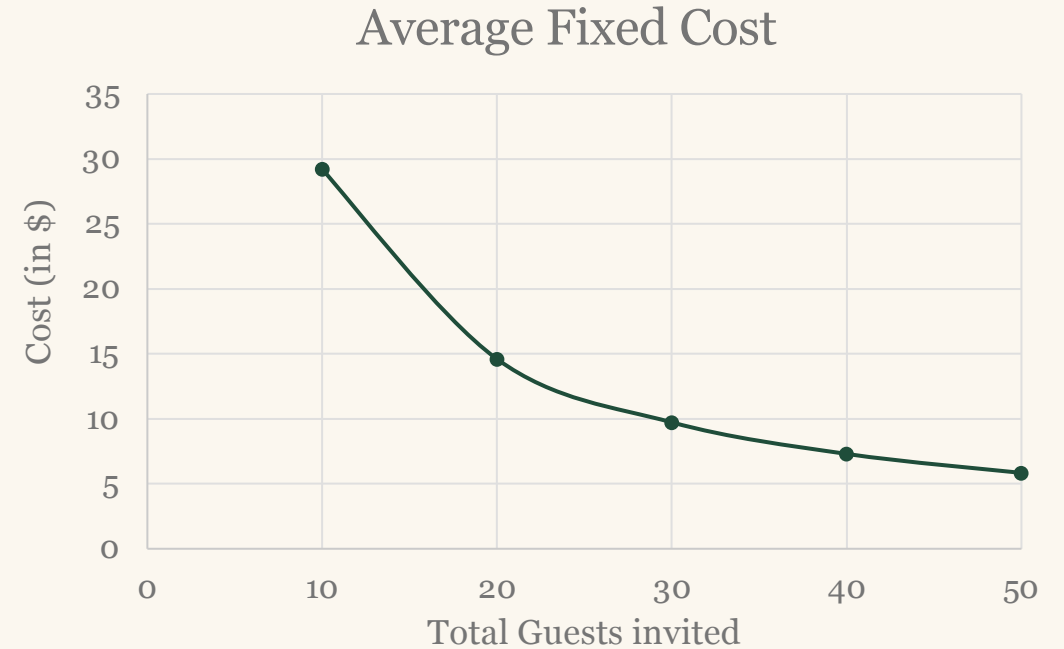
$$AFC = FC / Q$$

Marginal Falls Party:

$$\text{Average Fixed Cost} = \text{Park Rent} / \text{Number of Guests}$$

Average Fixed Cost

Guests	Fixed Cost	Average Fixed Cost
10	\$292	\$29.20
20	\$292	\$14.60
30	\$292	\$9.73
40	\$292	\$7.30
50	\$292	\$5.84



The fixed cost per guest keeps falling.

Not all Costs are Fixed

More guests also mean:

- more food
- more drinks
- more cleanup
- more crowding

Average Variable Cost

Remember:

$$TC = FC + VC$$

Average Variable Cost is Variable Cost divided by output.

$$AVC = VC / Q$$

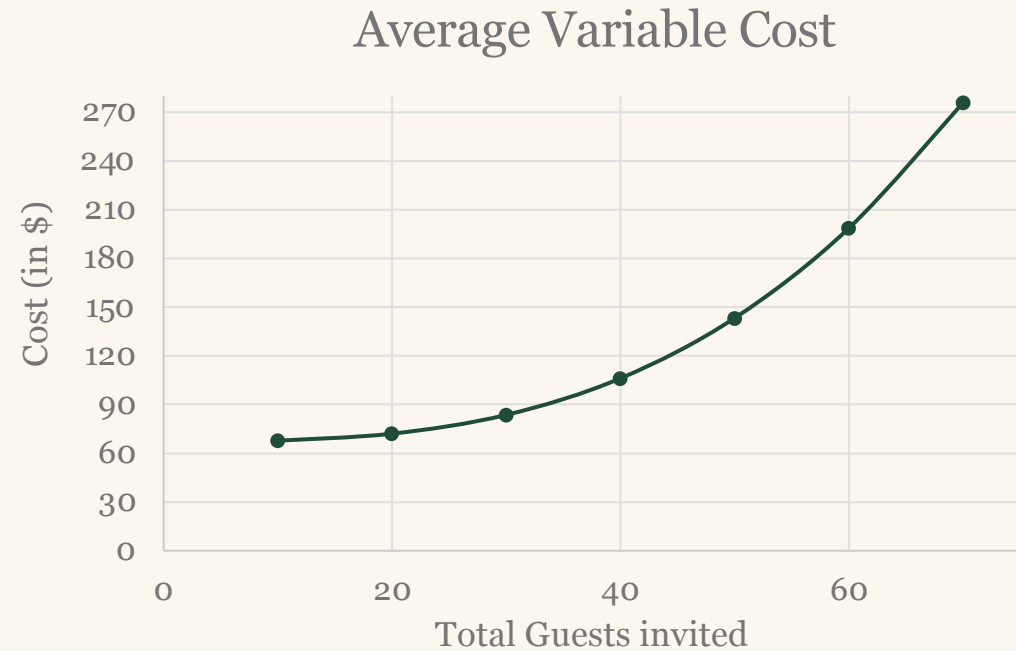
Marginal Falls Party:

Average Variable Cost = Guest-related cost / Number of Guests

(Food, drinks, plates, cleanup, and crowding cost per guest.)

Average Variable Cost

Guests	Variable Cost	Average Variable Cost
10	\$678	\$67.80
20	\$1,442	\$72.10
30	\$2,510	\$83.67
40	\$4,247	\$106.18
50	\$7,164	\$143.28



The variable cost per guest rises when the party becomes harder to manage.

The Cost Split Falls then Rises

$$\begin{aligned}\text{Cost Split per Guest} \\ &= \text{Setup Split per Guest} + \text{Guest-Related Cost per Guest}\end{aligned}$$

$$\text{ATC} = \text{AFC} + \text{AVC}$$

Guests	Setup Split (AFC)	Guest-Related Split (AVC)	Cost Split (ATC)
10	\$29.20	\$67.80	\$97.00
20	\$14.60	\$72.10	\$86.70
30	\$9.73	\$83.67	\$93.40
40	\$7.30	\$106.18	\$113.48

At first:

The rent is spread over more guests, so the cost split falls.

Later:

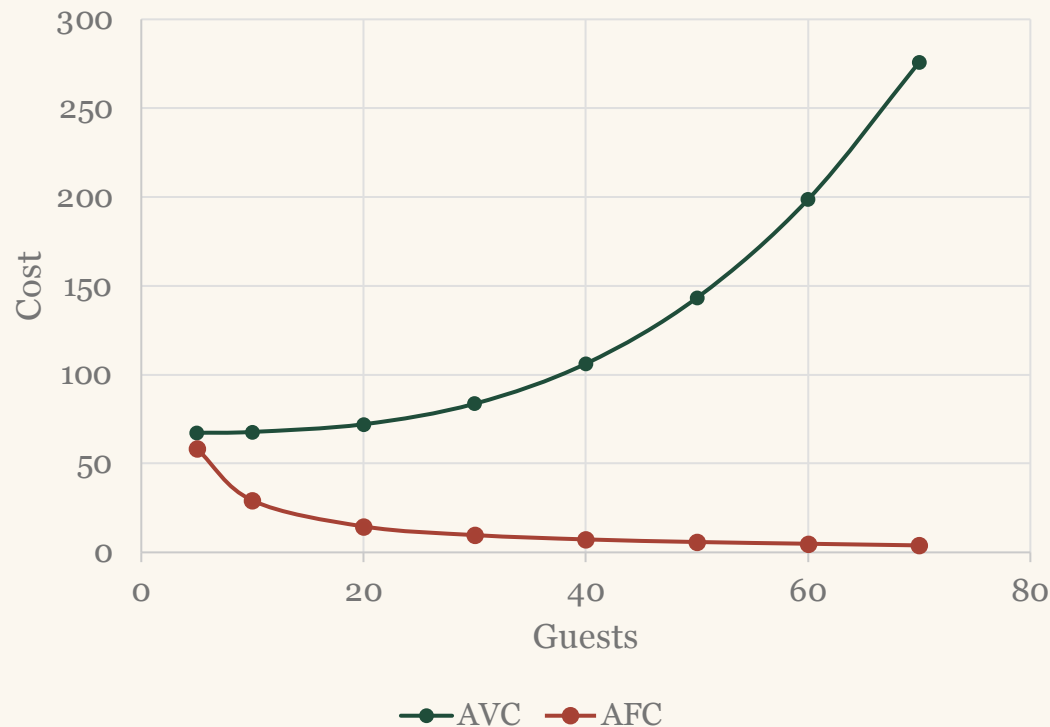
Food, drinks, cleanup, and crowding costs rise quickly, so the cost split rises.

That is why the Cost Split curve is U-shaped.

The Cost Split Falls then Rises

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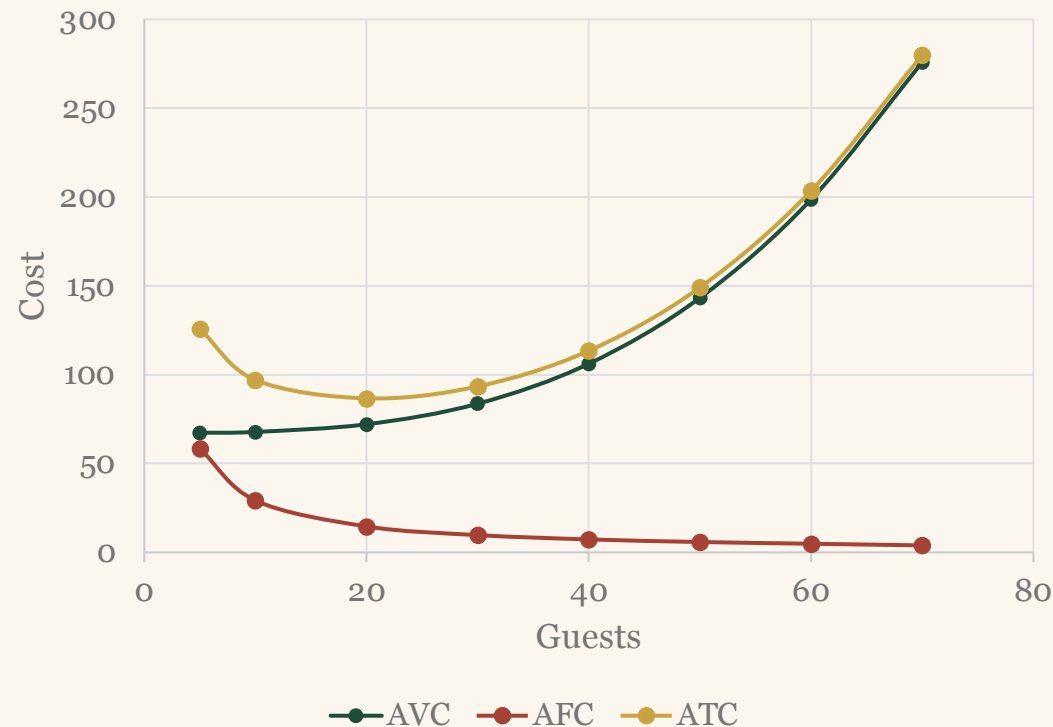
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The Cost Split Falls then Rises

Cost Split per Guest
= Setup Split per Guest + Guest-Related Cost per Guest

$$ATC = AFC + AVC$$



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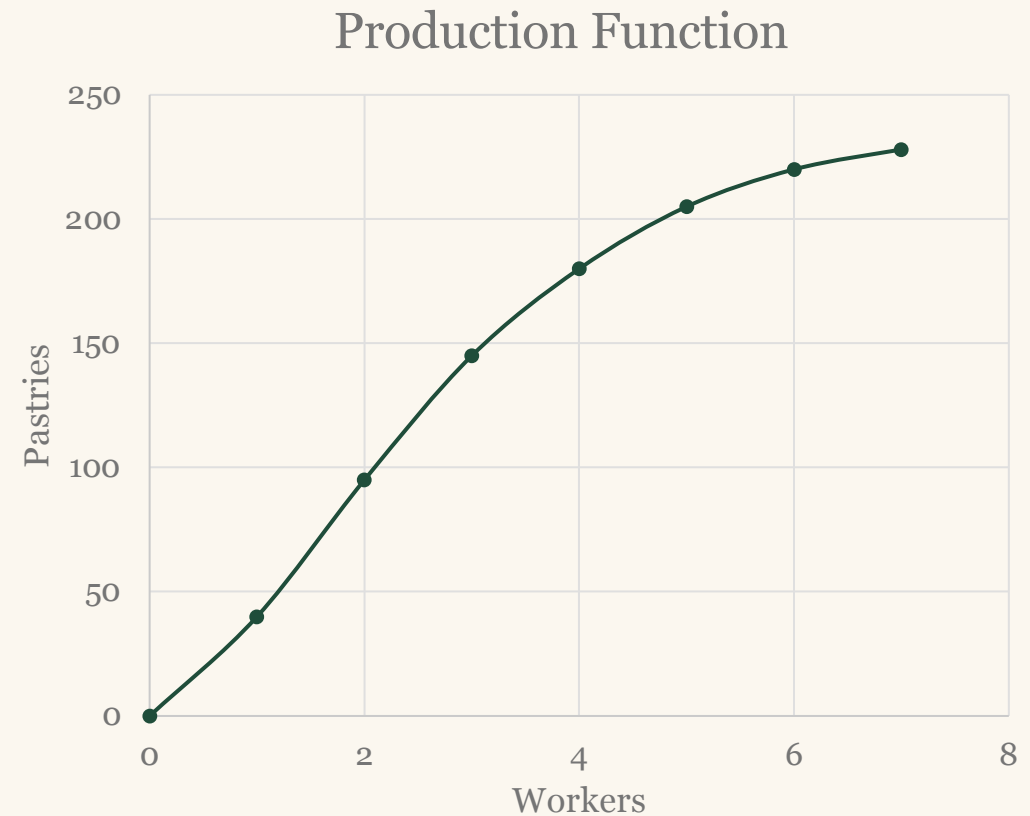
Production Function

A production function shows the relationship between inputs and the maximum output a firm can produce.



Bakery Production Ledger (Short Run)

Workers	Ovens	Pastries
0	2	0
1	2	40
2	2	95
3	2	145
4	2	180
5	2	205
6	2	220
7	2	228



Marginal Product of Labor

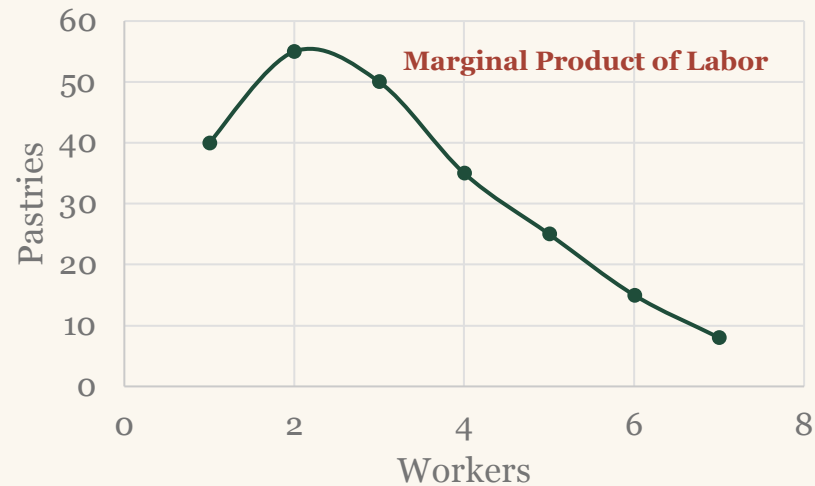
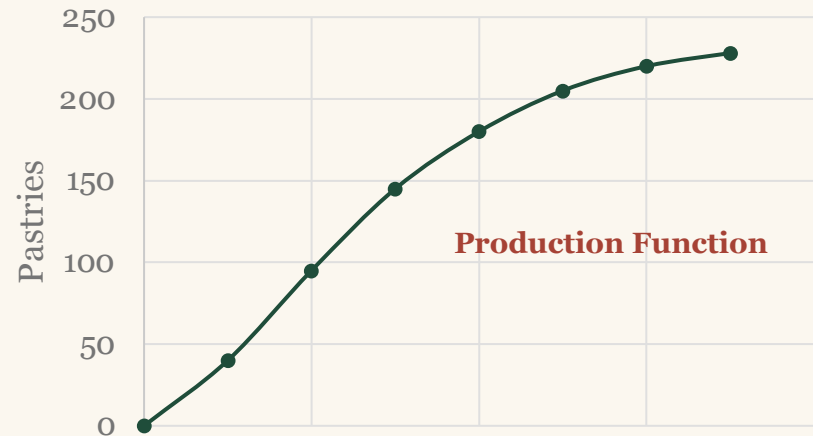
Marginal Product of Labor is the extra output produced by one more worker.

$MP_L = \text{Change in Output} / \text{Change in Labor}$

$$= \frac{\Delta Q}{\Delta L} = \frac{Q_2 - Q_1}{L_2 - L_1}$$

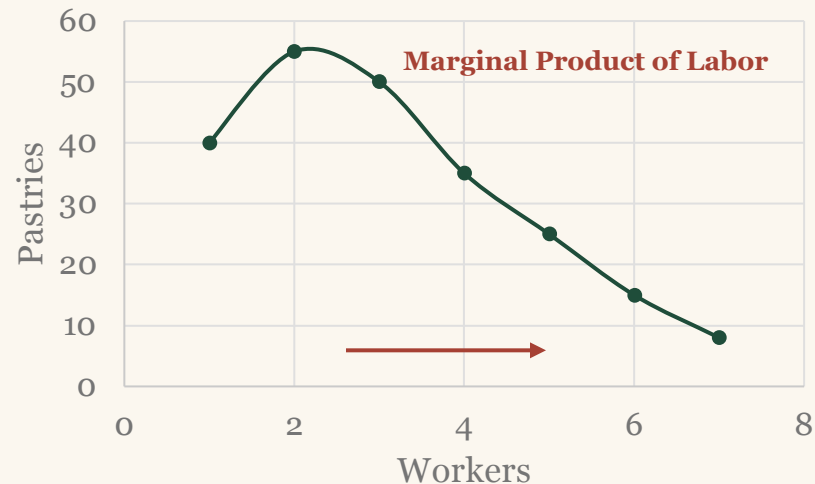
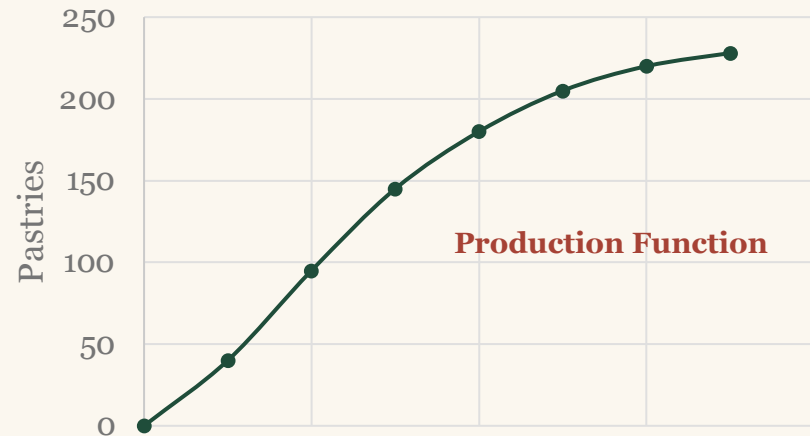
Workers (L)	Ovens (K)	Pastries (Q)	MP_L
0	2	0	–
1	2	40	40
2	2	95	55
3	2	145	50
4	2	180	35
5	2	205	25
6	2	220	15
7	2	228	8

Marginal Product of Labor



Workers	Ovens	Pastries	MP _L
0	2	0	—
1	2	40	40
2	2	95	55
3	2	145	50
4	2	180	35
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6	2	220	15
7	2	228	8

Marginal Product of Labor



Law of Diminishing Marginal Product of Labor:

At some point, adding more of a variable input to a fixed input causes marginal product to fall.

Average Product of Labor

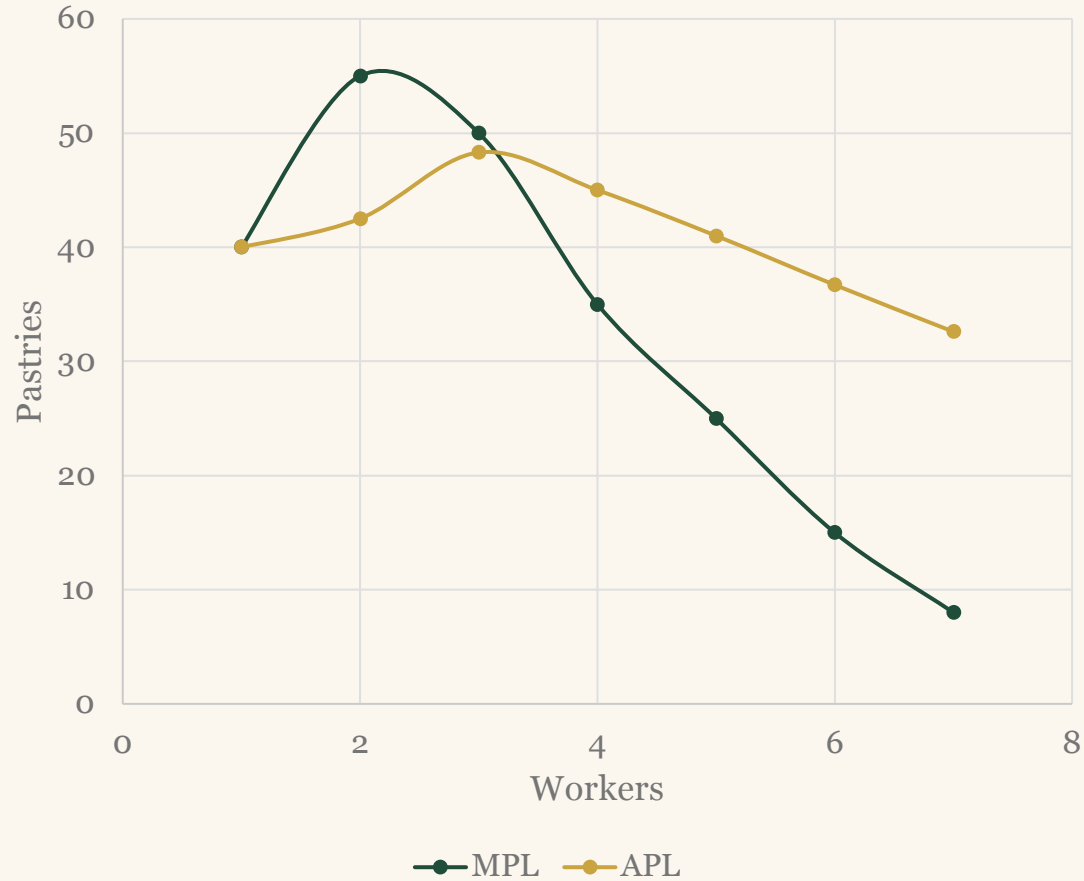
Average Product of Labor is output produced per worker.

$AP_L = \text{Total Output} / \text{Total Labor}$

$$= \frac{Q}{L}$$

Workers (L)	Ovens (K)	Pastries (Q)	AP_L
0	2	0	–
1	2	40	40
2	2	95	42.5
3	2	145	48.3
4	2	180	45
5	2	205	41
6	2	220	36.7
7	2	228	32.6

Average Product of Labor



Workers (L)	Ovens (K)	Pastries (Q)	AP_L	MP_L
0	2	0	—	—
1	2	40	40	40
2	2	95	42.5	55
3	2	145	48.3	50
4	2	180	45	35
5	2	205	41	25
6	2	220	36.7	15
7	2	228	32.6	8

From Productivity to Cost

The bakery pays wage to each worker, which is the cost of labor.

- If one more worker produces many pastries, each extra pastry is cheap.
- If one more worker produces few pastries, each extra pastry is expensive.

Marginal Productivity of Labor & Marginal Cost

Marginal Cost

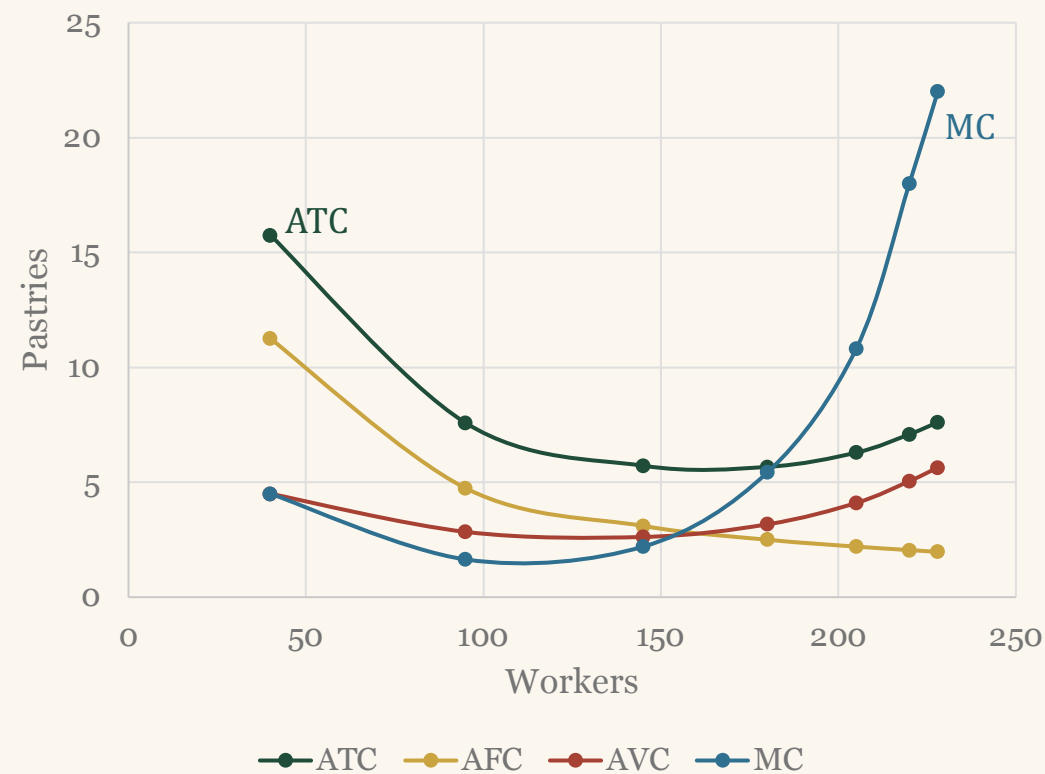
Marginal Cost is the change in total cost from producing one more unit.

MC = Change in Total Cost / Change in Output

$$= \frac{\Delta TC}{\Delta Q} = \frac{TC_2 - TC_1}{Q_2 - Q_1}$$

Workers (L)	Pastries (Q)	Total Cost (TC)	MP _L	MC
1	40	\$630	40	\$4.50
2	95	\$720	55	\$1.64
3	145	\$830	50	\$2.20
4	180	\$1,020	35	\$5.43
5	205	\$1,290	25	\$10.80
6	220	\$1,560	15	\$18.00
7	228	\$1,736	8	\$22.00

Marginal Cost



Workers (L)	Output (Q)	TC	VC	MC	ATC	AFC	AVC
1	40	\$630	\$180	\$4.50	\$15.75	\$11.25	\$4.50
2	95	\$720	\$270	\$1.64	\$7.58	\$4.74	\$2.84
3	145	\$830	\$380	\$2.20	\$5.72	\$3.10	\$2.62
4	180	\$1,020	\$570	\$5.43	\$5.67	\$2.50	\$3.17
5	205	\$1,290	\$840	\$10.80	\$6.29	\$2.20	\$4.10
6	220	\$1,560	\$1,110	\$18.00	\$7.09	\$2.05	\$5.05
7	228	\$1,736	\$1,286	\$22.00	\$7.61	\$1.97	\$5.64

Costs in the Long Run

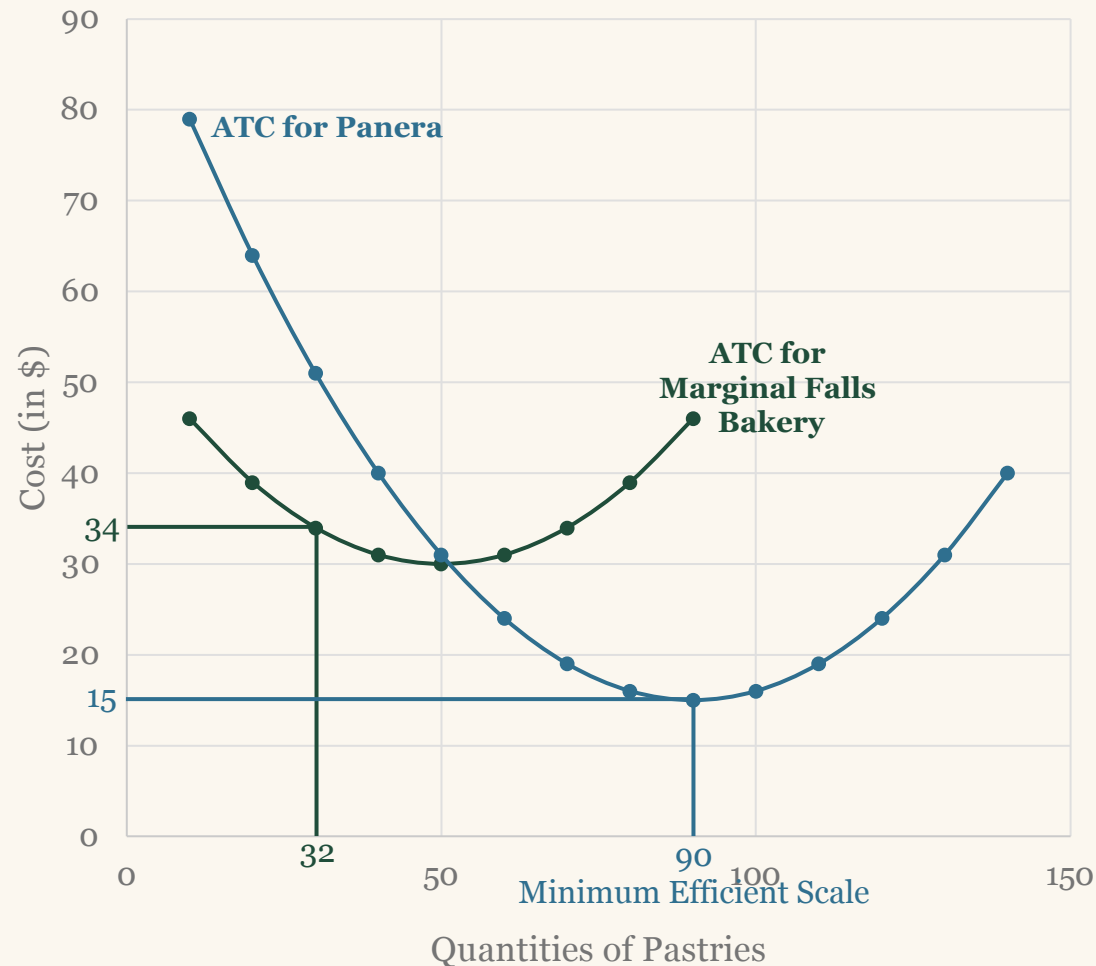
In the short run, the bakery is **stuck with 2 ovens**.

In the long run, the bakery can change everything:

- bigger kitchen, more ovens, better layout, more workers, new delivery system.

“What size bakery should we build?”

Economies of Scale



Marginal Falls bakery: Low setup cost.
Works well at low output.

Panera: High setup cost. Works well only at
high output.

Panera becomes cheaper: Bulk buying,
Specialized workers, Better equipment.

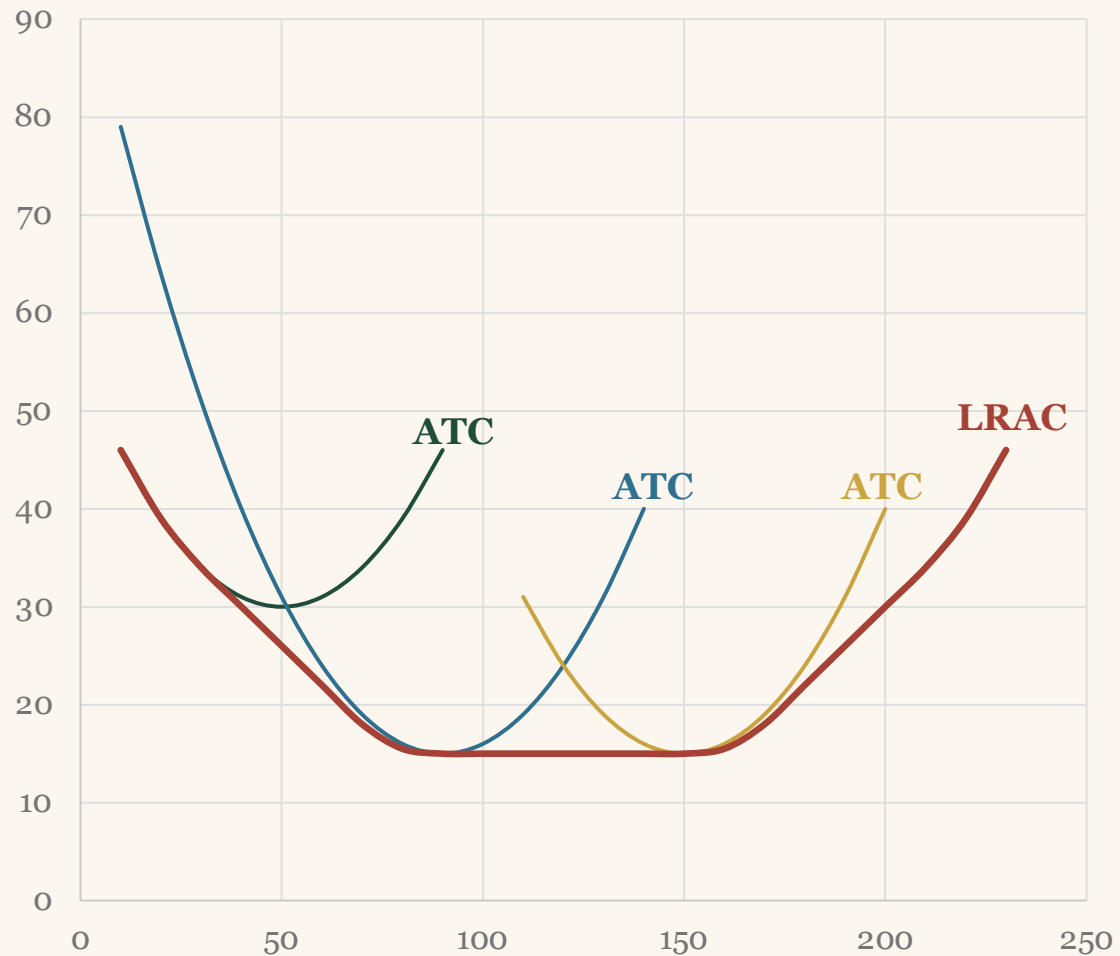
Economies of scale:

Average cost falls as output rises.

Minimum efficient scale:

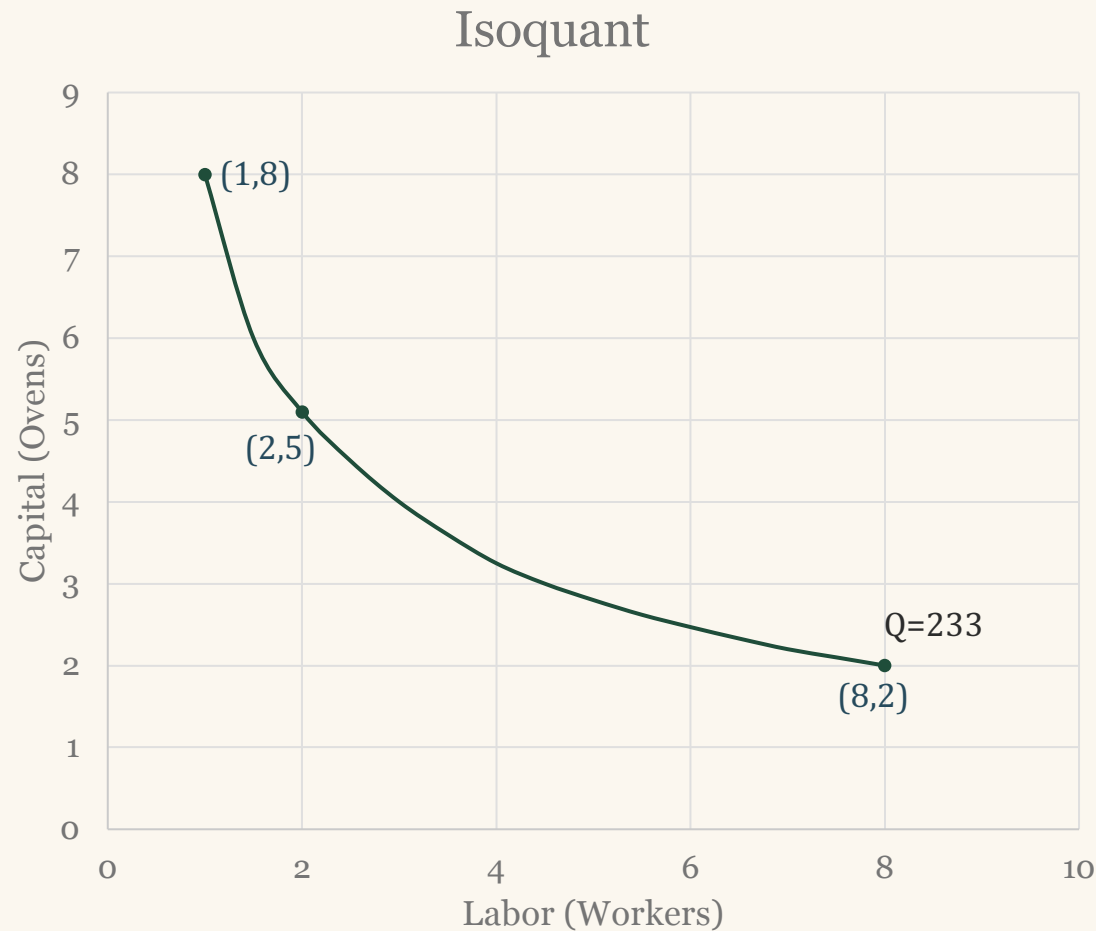
The lowest output where ATC reaches its
minimum.

Long Run Average Cost Curve



Long-run average cost shows the lowest possible cost per pastry when the bakery can choose all inputs.

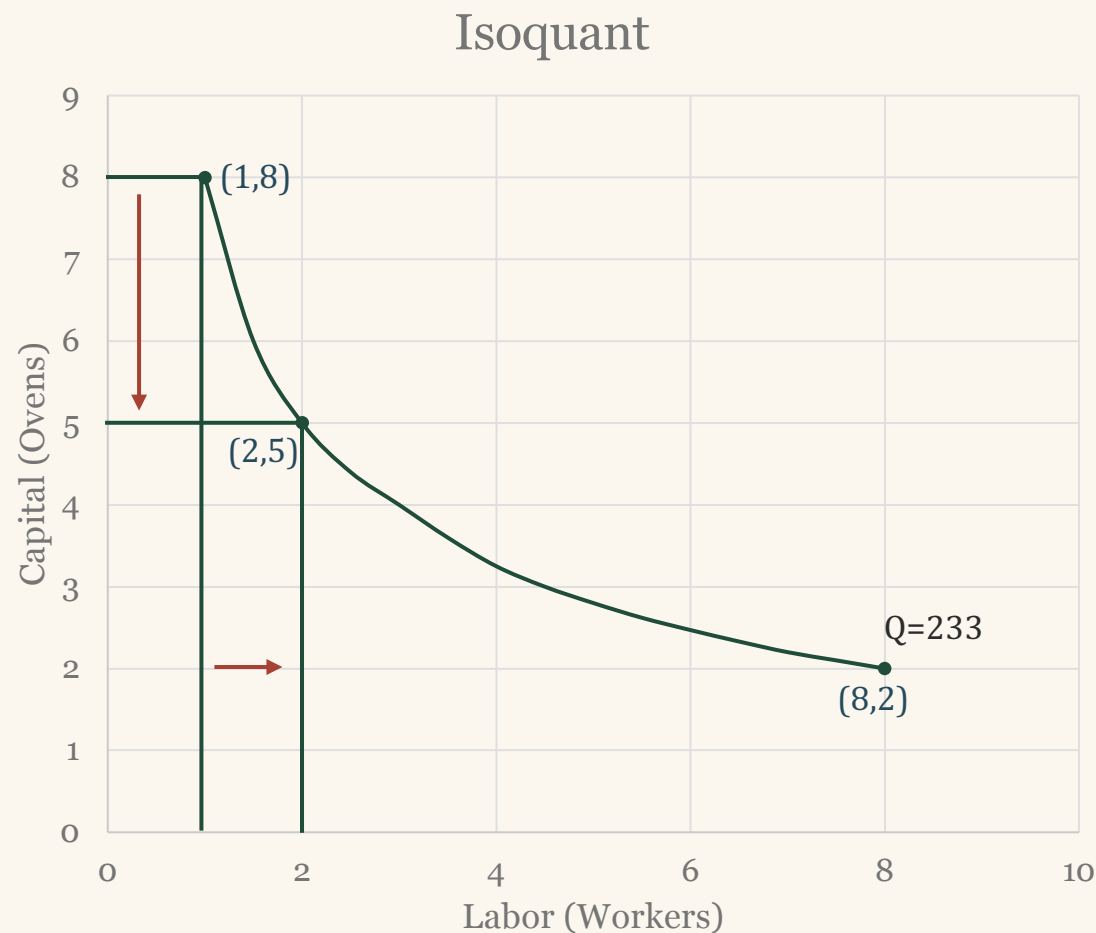
Same Production, Different Inputs



Isoquants:

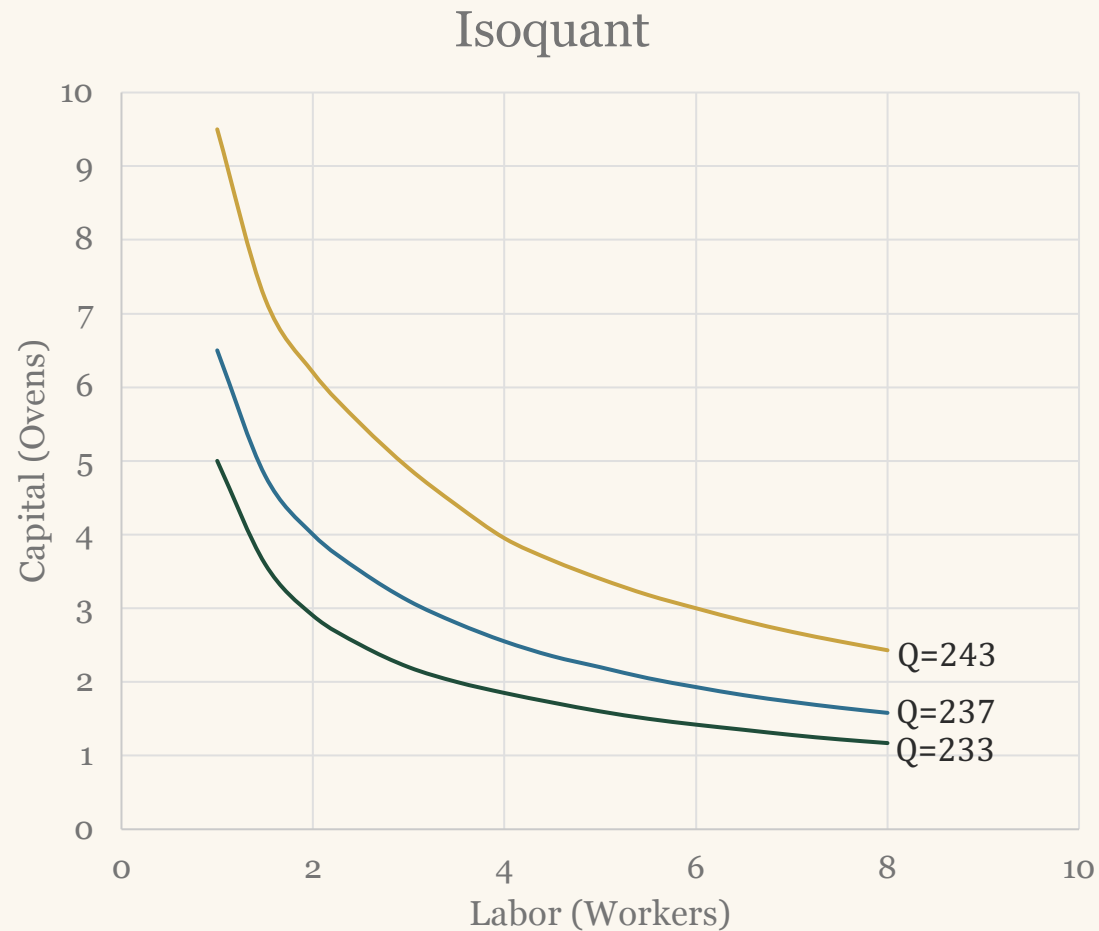
An isoquant shows all combinations of inputs that produce the same output.

Isoquants



- The slope shows how the bakery can trade ovens for workers while keeping output the same.
- This is called the **Marginal Rate of Technical Substitution**.
- $MRTS$ = how many ovens can be replaced by one more worker.

Isoquants



Isoquants:
Higher isoquants require more inputs and produce more output.

Isocost Line

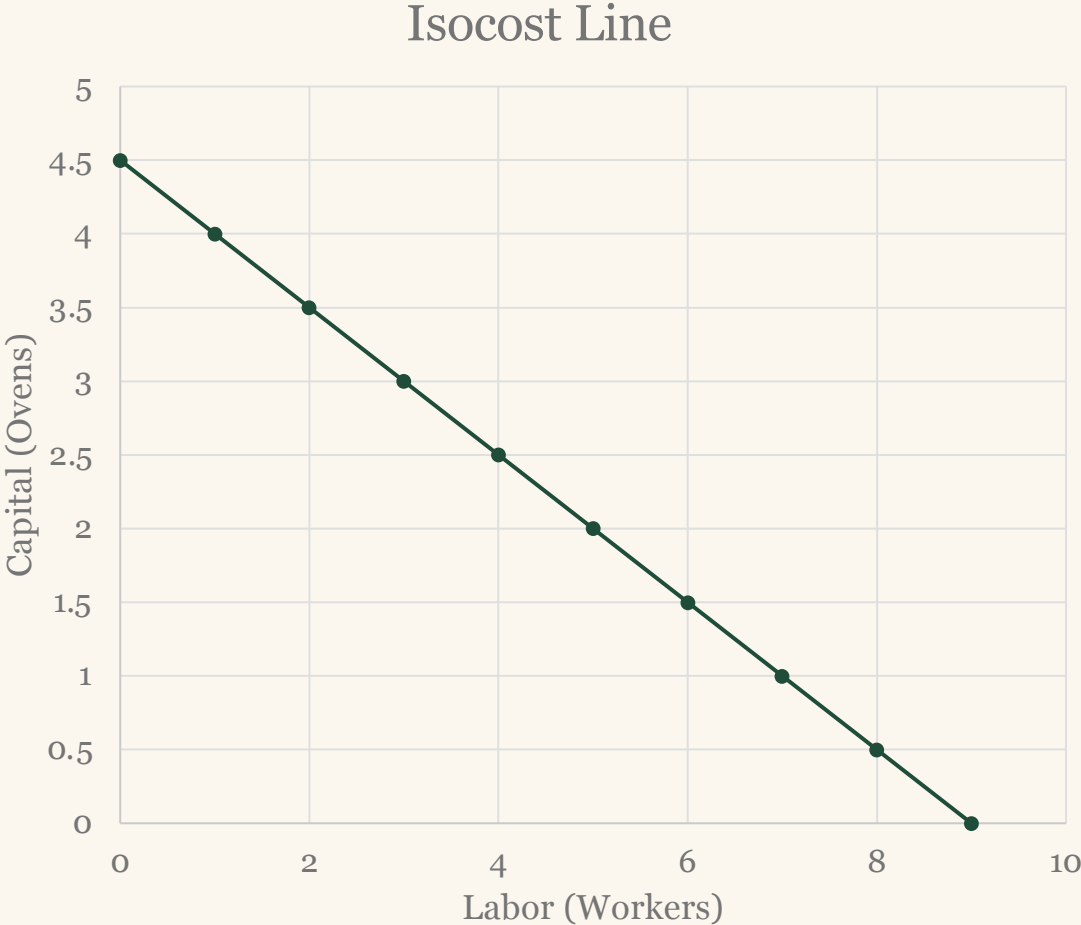
Capital (Ovens)	Labor (Workers)	Total Cost
4	1	\$900
3	3	\$900
2	5	\$900
1	7	\$900

An **isocost line** shows all combinations of inputs with the same total cost.

Suppose, wage = \$100, and oven rental cost = \$200

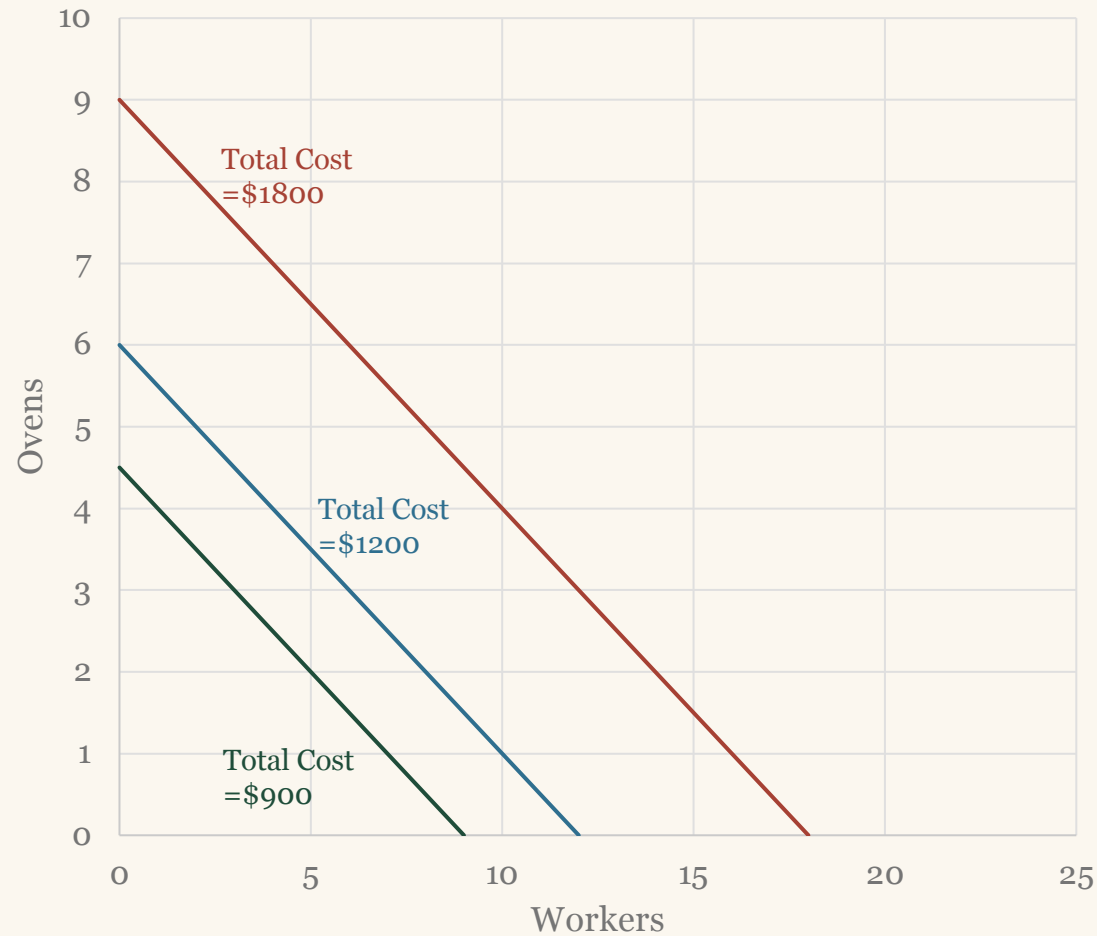
And Total Cost = \$900

Isocost Line



Capital (Ovens)	Labor (Workers)	Total Cost
4	1	\$900
3	3	\$900
2	5	\$900
1	7	\$900

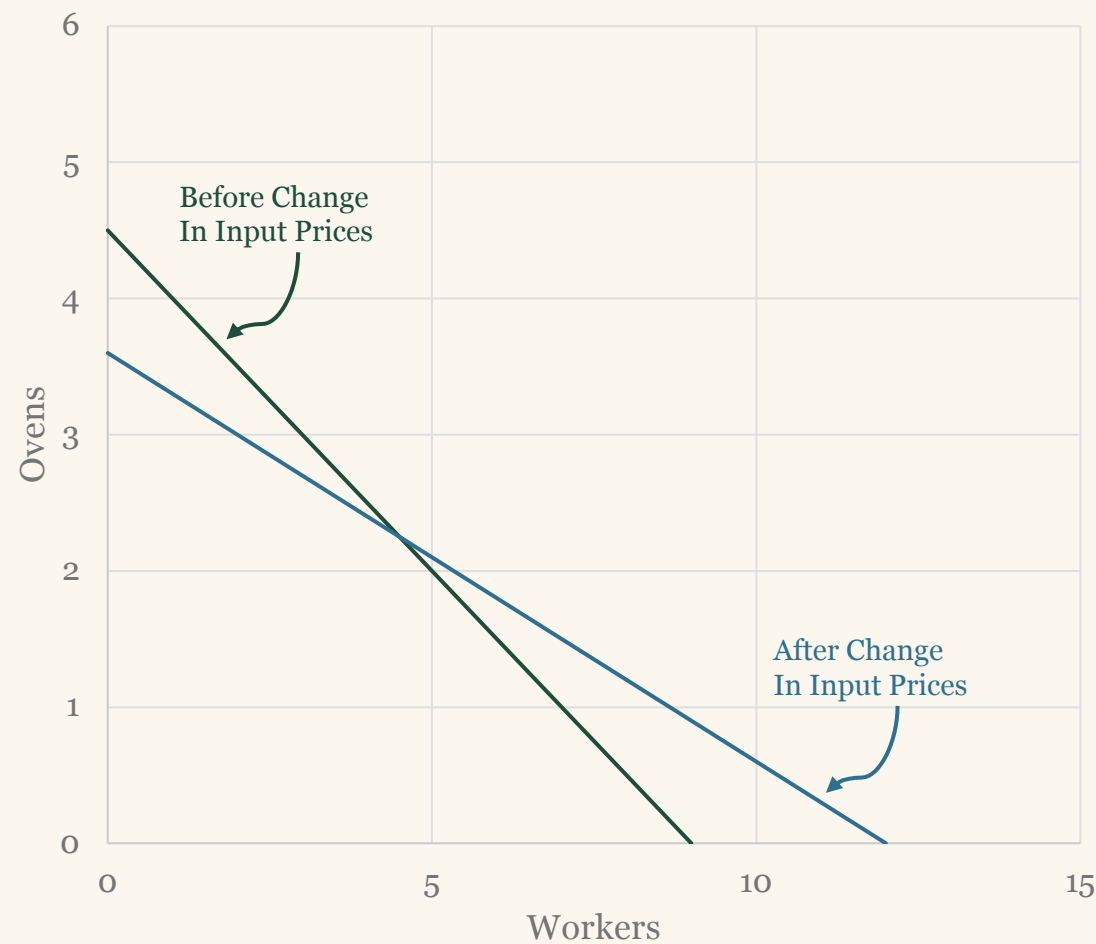
Isocost Line



Higher isocost lines mean higher total cost.

Same slope means input prices have not changed.

Change in Input Prices



Isocost Line:

$$100*L + 200*K = \$900$$

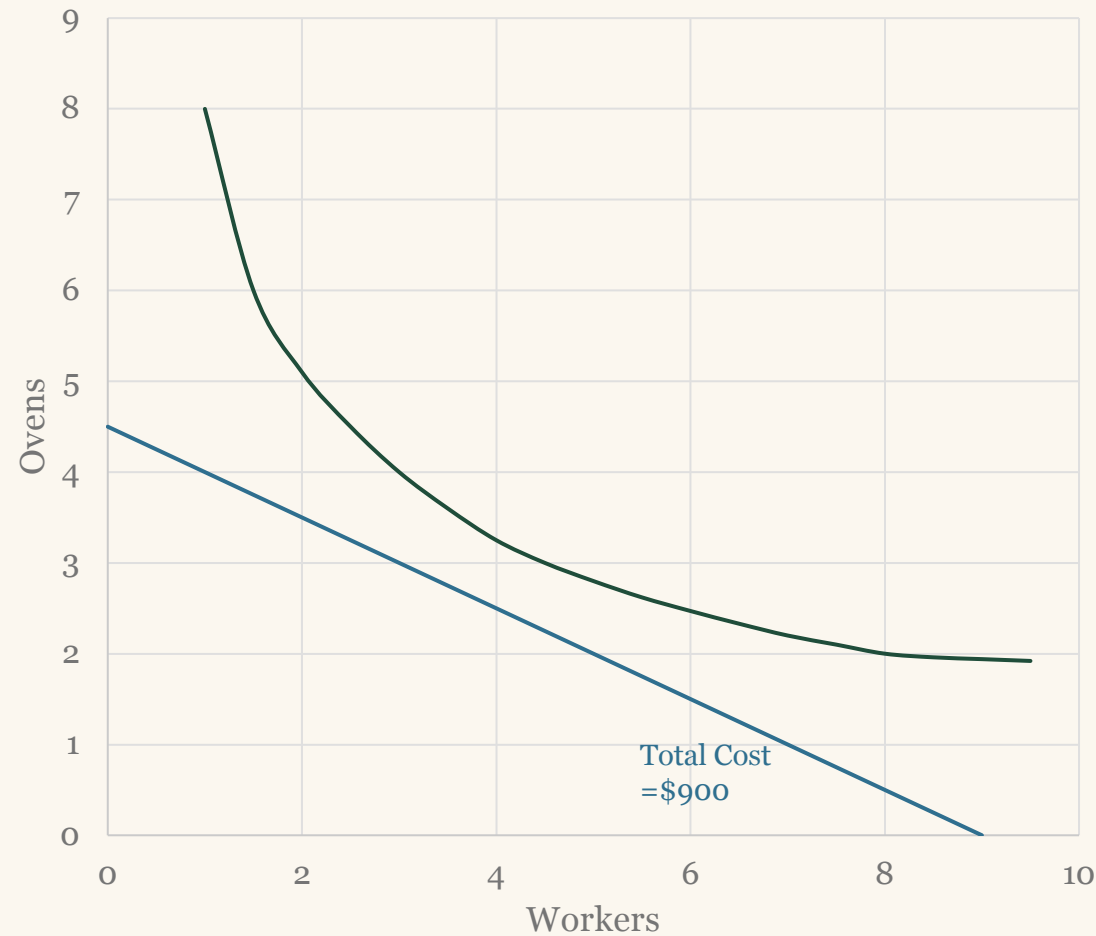
Labor wage is \$75

Oven rental is \$250

$$75*L + 250*K = \$900$$

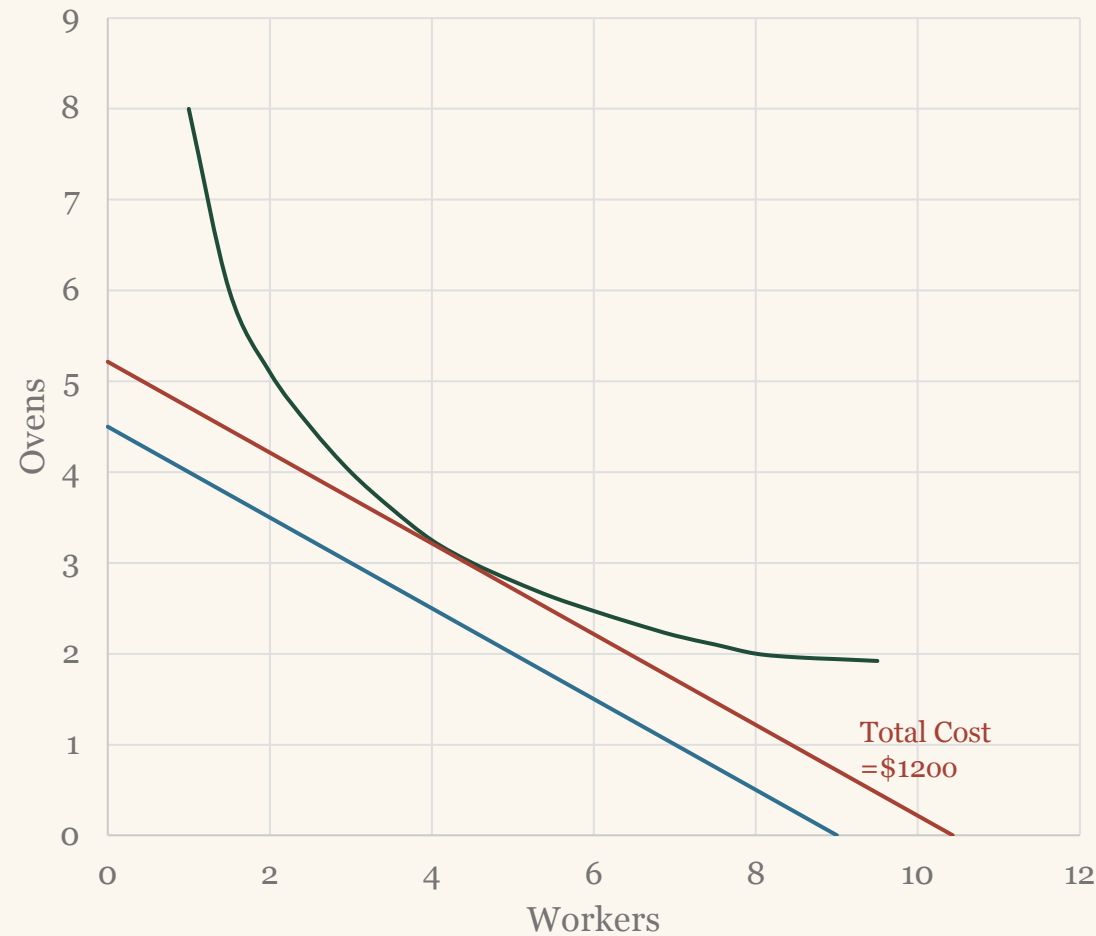
Slope of Isocost Line **changes**.

Cost-Minimizing Input Choice



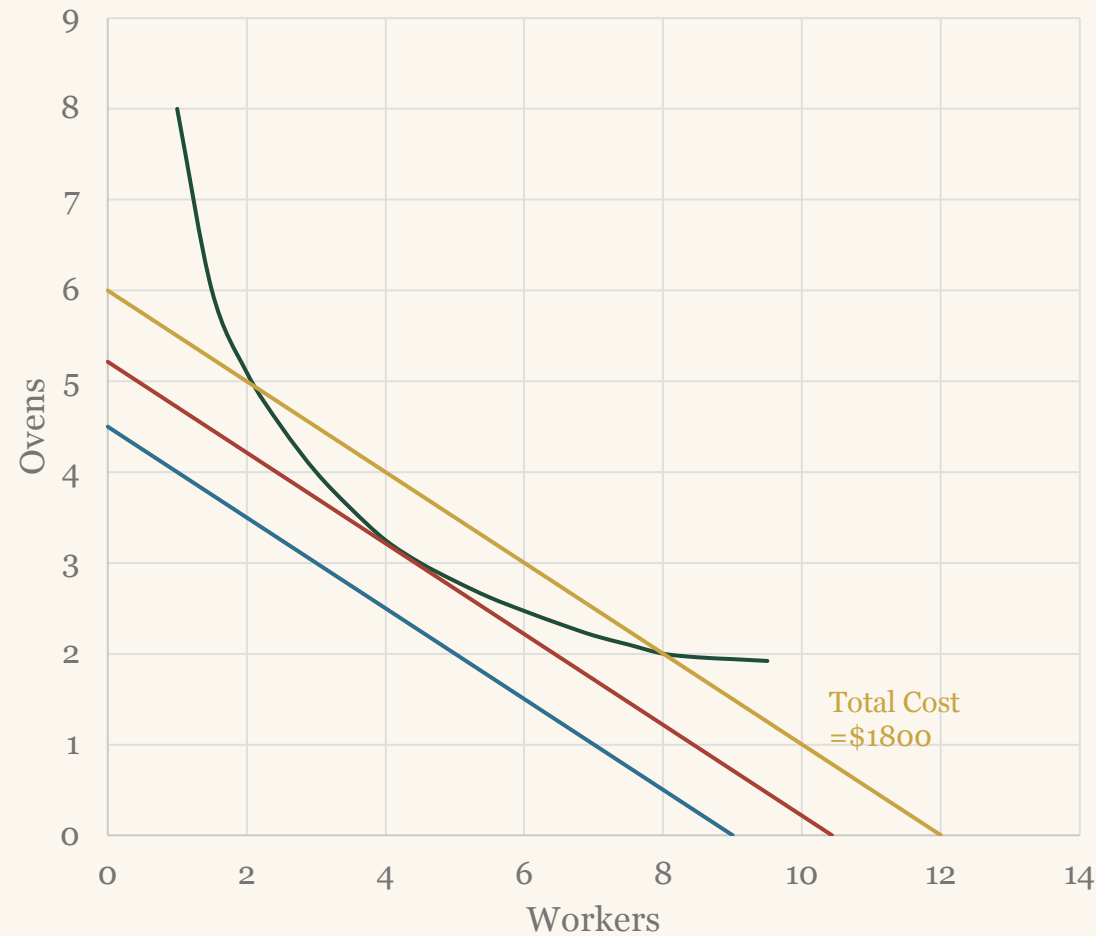
The Bakery wants to produce 233 pastries at the lowest possible cost.

Cost-Minimizing Input Choice



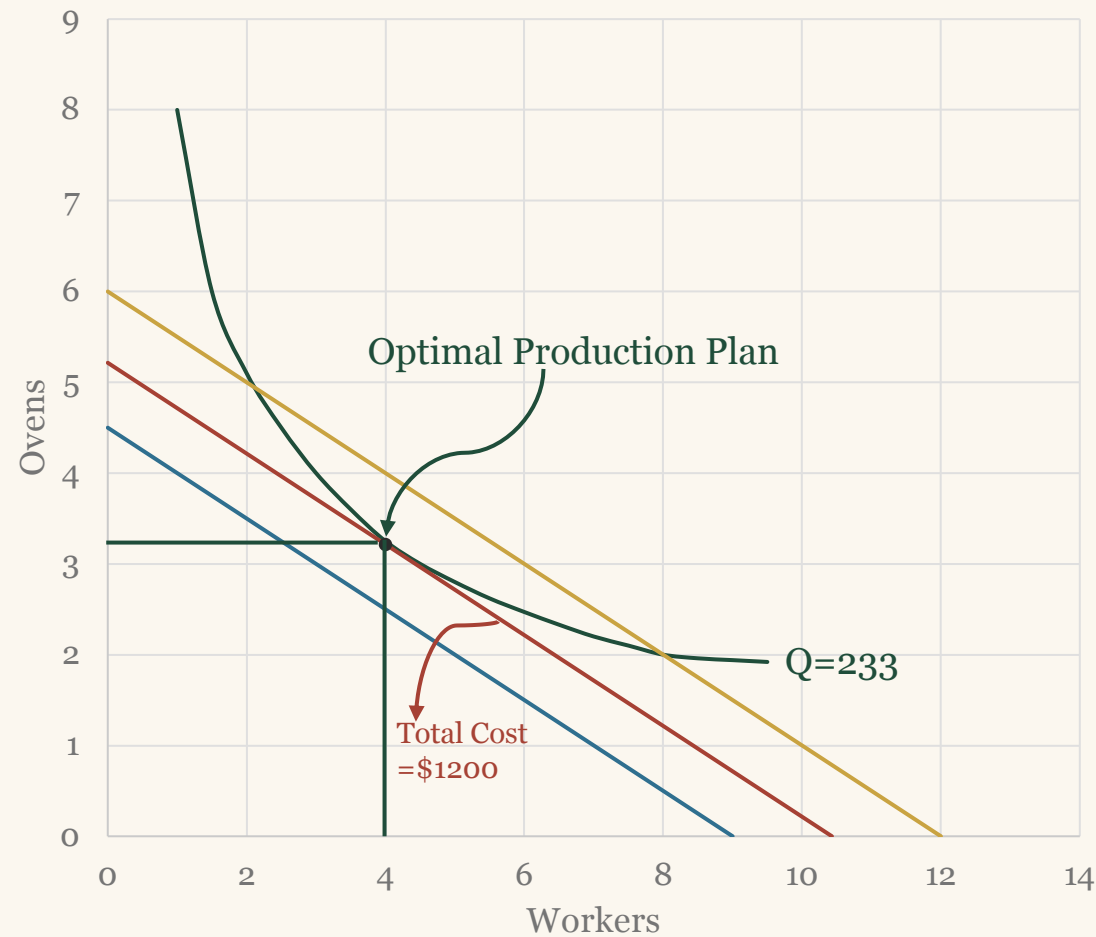
The Bakery wants to produce 233 pastries at the lowest possible cost.

Cost-Minimizing Input Choice



The Bakery wants to produce 233 pastries at the lowest possible cost.

Cost-Minimizing Input Choice



The Bakery wants to produce 233 pastries at the lowest possible cost.

Choose the lowest isocost line that touches the $Q = 233$ isoquant.

Coming Up

Optimal choice at different prices → Demand Curve

Optimal production plan at different input prices → ?

Counterpart of Demand Curve for Producers.